8TH GRADE MATH • SUPPORT NOTES

Unit 1: The Real Number System

Standards NC.8.NS.1 & NC.8.NS.2 • Guided Notes Built for You

Name: ______________________ Date: ______________________

Class: ______________________ Period: ______________________

How to Use These Notes

- **Words to Know** boxes give you the meaning of each new word in plain language.
- **Remember** boxes hold the one big idea. If you forget everything else, keep this.
- **Steps** boxes break a problem into small pieces. Do one line at a time.
- **I Do** shows a finished example. **You Try** is your turn — go slow.
- **Watch Out** warns you about the mistake most people make here.
- Blanks like this ______ are for you to fill in with your teacher.

What Is In This Packet

- **Warm-Up:** the older skills we need first (signs, fractions, the number line).
- **Part 1 — Guided Notes:** sections 1.1 through 1.6.
- **Part 2 — Practice:** fewer problems, more help on each one.
- **Part 3 — Answer Key:** check your work after you try.

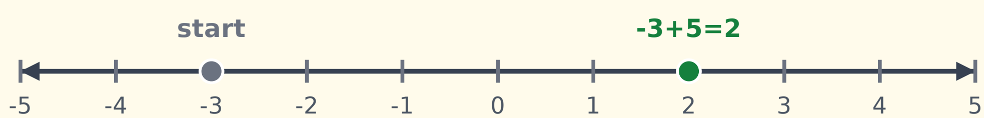
★ Warm-Up: Skills We Need First

Before new math, let's wake up the math we already have. These come back all unit.

⚠ Sign Rules (positive & negative)

When you…	Same signs	Different signs
Multiply or Divide	answer is **positive (+)** (−)(−)=+	answer is **negative (−)** (−)(+)=−
Add	add, keep the sign	subtract, keep the sign of the *bigger* number

Adding on a number line: **+ moves right, − moves left.**



Start at -3, move 5 right, land on 2

Fraction → Decimal

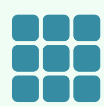
Top ÷ bottom. The fraction bar means "divide."

`3/4 means 3 ÷ 4 = 0.75`

`1/8 means 1 ÷ 8 = 0.125`

What Does "Squared" Mean?

x^2 means a number times itself: a real **square**.



3 × 3 = 9

Words to Know

Place value — the spot a digit sits in (ones, tenths, hundredths…)

Number line — a line where numbers grow as you go right

Inverse — the opposite operation that undoes another (+ undoes −)

1.1 Converting Numbers & Repeating Decimals

NC.8.NS.1

Words to Know

Terminating decimal — a decimal that **stops** (0.75)

Repeating decimal — a decimal where digits **repeat forever** (0.333...)

Bar notation — a bar over the digits that repeat: $0.\overline{3}$

Turn a Fraction Into a Decimal

Divide the **top** by the **bottom**. Fill in the answers with your teacher:

Fraction	Divide	Decimal
3/4	3 ÷ 4	______
1/8	1 ÷ 8	______
1/3	1 ÷ 3	______ (repeats!)

The Big Idea

A repeating decimal is **RATIONAL** — it can become a fraction.

A decimal that never repeats and never stops (like π) is **IRRATIONAL**.

Move the Decimal (percent & fraction)

Change	Rule	Try it
Decimal → Percent	move 2 places **right**	0.45 → ______
Percent → Decimal	move 2 places **left**	75% → ______
Decimal → Fraction	read it, write it, simplify	0.75 = 75/100 = ______

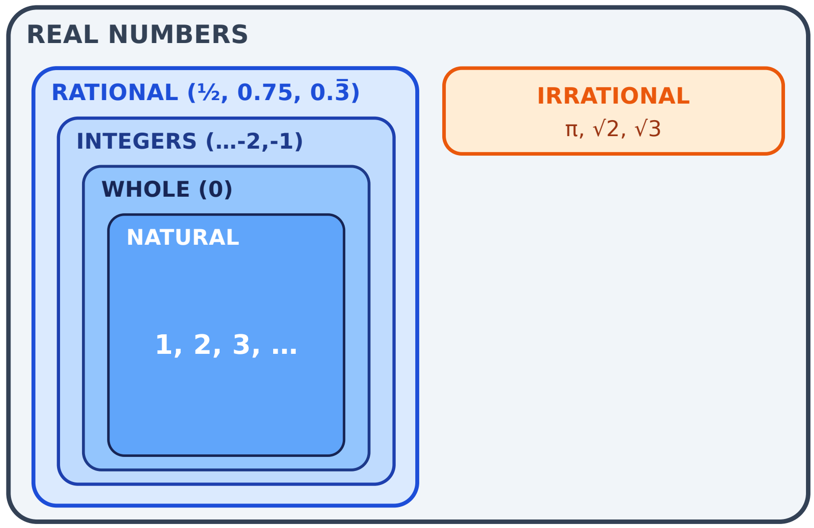
1.2 Number Classification

NC.8.NS.1

Every number lives in one or more “family” groups. Bigger groups hold the smaller ones.

Fill In Each Family

Family	Examples	Means…
Natural	1,2,3…	______ numbers
Whole	0,1,2,3…	naturals + ______
Integer	…−2,−1,0,1…	wholes + ______
Rational	½, 0.75, $0.\overline{3}$	can be a ______
Irrational	π, √2	______, ______ decimal



Sentence Frame for Answers

Say it the same way every time:

“_____ is **rational** because it can be written as a fraction.”

“_____ is **irrational** because its decimal never stops and never repeats.”

1.3 Perfect Squares & Square Roots

NC.8.NS.2

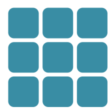
★ They Undo Each Other

Squaring and square-rooting are **inverse** (opposite) operations.

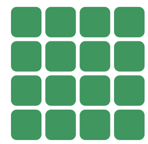
$$\text{If } 5^2 = 25\text{, then } \sqrt{25} = 5.$$

Know one → you know the other.

$2 \times 2 = 4$



$3 \times 3 = 9$



$4 \times 4 = 16$

Memorize These (fill them in)

n	3	4	5	6	7
n^2	____	____	____	____	____

n	8	9	10	11	12
n^2	____	____	____	____	____

⚠ Watch Out

$\sqrt{25} = 5$ is a **whole number**, so it is **rational**.

$\sqrt{10}$ is **not** a perfect square, so it is **irrational**.

1.4 Repeating Decimals → Fractions (the 9's Rule)

NC.8.NS.1

The 9's Rule

The repeating digits go on **top**. The bottom is all **9's** — one 9 for each repeating digit.

Build the Fraction

# repeating digits	Bottom	Example	Simplified
1 digit	______	$0.\overline{3} = 3/9$	______
2 digits	______	$0.2\overline{7} = 27/99$	______
3 digits	______	$0.12\overline{5} = 125/999$	______

I Do: change $0.2\overline{7}$ into a fraction

1. Repeating digits: **27** → that is **2** digits.
2. Bottom = two 9's = **99**.
3. Fraction = **27/99**.
4. Simplify: 27/99 = **3/11**.

Watch Out

The 9's rule only works when the repeat starts **right after** the decimal point. Always simplify your final fraction.

1.5 Estimating Irrational Square Roots

NC.8.NS.2

★ Big Idea

Most square roots are messy (irrational). We **estimate** — find the two whole numbers it sits between.

Method 1 — Use a Number Line

1. Find the perfect square just **below** and just **above** your number.
2. Take the square root of each. Those are your two whole numbers.
3. Decide: is your number closer to the lower or higher one?
4. Place it on the line.

√16√20√2520

is between 16 and 25, so $\sqrt{20}$ is between 4 and 5

Method 2 — 4-Step Math: estimate $\sqrt{45}$

1. Squares around it: $36 < 45 < 49 \rightarrow \sqrt{36}=6$, $\sqrt{49}=7$.
2. Make a fraction: $(45 - 36) \div (49 - 36) = 9 \div 13$.
3. $9 \div 13 \approx 0.69$.
4. $6 + 0.69 =$ **6.7** $\rightarrow \sqrt{45} \approx 6.7$

Formula: $\sqrt{n} \approx \text{lower root} + (n - \text{lower}^2) \div (\text{upper}^2 - \text{lower}^2)$

1.6 Ordering Rational & Irrational Numbers

NC.8.NS.2

★ One Trick Fixes Everything

Turn **every** number into a decimal first. Then they are easy to compare.

I Do: order √5, 2¼, 7/3, 2.1 (least to greatest)

Number	As a decimal	How
√5	≈ 2.24	4-step estimate
2¼	2.25	2 + (1÷4)
7/3	≈ 2.33	7 ÷ 3
2.1	2.10	already a decimal

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Order: 2.1 < √5 < 2¼ < 7/3
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⚠ Watch Out

Line up the same number of decimal places. 3.5 is bigger than 3.375 even though “375” looks long.

Estimate irrational roots **before** you place them.

Part 2 Practice — Classify Numbers

Use the sentence frame from 1.2. Tip: a square root is rational only if it is a perfect square.

Reminder

"__ is rational because it can be a fraction."

"__ is irrational because its decimal never stops/repeats."

#	Number	Rational or Irrational?	Why?
1	0.333...		
2	π		
3	√49		
4	√10		
5	−3/8		
6	0.1212...		
7	√144		
8	2/7		

Part 2 Practice — Estimate & Order

Estimate each root to the nearest tenth (use the 4 steps)

#	Root	Between which squares?	Estimate
1	$\sqrt{5}$		
2	$\sqrt{20}$		
3	$\sqrt{30}$		
4	$\sqrt{50}$		
5	$\sqrt{8}$		

Order from least to greatest (convert first!)

1. π, $\sqrt{10}$, 10/3, 3.5

Decimals: ______ ______ ______ ______

Order: ______ < ______ < ______ < ______

2. 1.5, $\sqrt{2}$, 7/4, $\sqrt{3}$

Decimals: ______ ______ ______ ______

Order: ______ < ______ < ______ < ______

★ ANSWER KEY ★

For checking your work after you try. Teachers: hand out after the lesson.

Key Guided Notes Answers

1.1 3/4=**0.75**; 1/8=**0.125**; 1/3=**$0.\overline{3}$**. Repeating=**rational**; never-ending=**irrational**. 0.45→**45%**; 75%→**0.75**; 0.75=**3/4**.

1.2 Natural=**counting**; Whole=naturals+**zero**; Integer=wholes+**negatives**; Rational=can be a **fraction**; Irrational=**non-repeating, non-terminating** decimal.

1.3 3^2=**9**, 4^2=**16**, 5^2=**25**, 6^2=**36**, 7^2=**49**, 8^2=**64**, 9^2=**81**, 10^2=**100**, 11^2=**121**, 12^2=**144**.

1.4 Bottoms: **9, 99, 999**. 3/9=**1/3**; 27/99=**3/11**; 125/999 (already simplest).

1.5 $\sqrt{45} \approx$ **6.7** (between $\sqrt{36}=6$ and $\sqrt{49}=7$).

1.6 $2.1 < \sqrt{5} < 2\frac{1}{4} < 7/3$.

Key Practice Answers

Part A — Classify

1. **Rational** ($0.\overline{3}=1/3$) 2. **Irrational** (π) 3. **Rational** ($\sqrt{49}=7$) 4. **Irrational** 5. **Rational** 6. **Rational** (repeats) 7. **Rational** ($\sqrt{144}=12$) 8. **Rational**

Part B — Estimates (nearest tenth)

$\sqrt{5}\approx$**2.2**, $\sqrt{20}\approx$**4.4**, $\sqrt{30}\approx$**5.5**, $\sqrt{50}\approx$**7.1**, $\sqrt{8}\approx$**2.8**

Part C — Order

1. 3.14(π) $\approx$3.14, $\sqrt{10}\approx$3.16, $10/3\approx$3.33, 3.5 → **$\pi < \sqrt{10} < 10/3 < 3.5$**

2. 1.5, $\sqrt{2}\approx$1.41, $7/4=$1.75, $\sqrt{3}\approx$1.73 → **$\sqrt{2} < 1.5 < \sqrt{3} < 7/4$**

End of Unit 1 support packet.